Design of Beam-Splitter *

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Introduction

The objective of this work is to design, simulate, and layout a silicon photonic four-port beam splitter suitable for operation in the telecom wavelength range. The design flow starts from the photonic platform assumptions, proceeds through single-waveguide and coupled-waveguide modal analysis, and ends with an automated GDS layout generation and electromagnetic verification workflow. The final goal is to obtain a set of directional couplers with different gaps and coupling lengths, identify the geometries that provide near-50/50 splitting, and integrate them into a testable chip layout with grating-coupler access structures.

We first define the assumed fabrication platform and material stack. The nominal platform is a silicon-on-insulator process with a silicon device layer of thickness

$$t_{\mathrm{Si}} = 220 \text{ nm},$$

a buried oxide layer, and silicon dioxide upper cladding. The operating wavelength range is chosen to be approximately

$$1500 \text{ nm} \leq \lambda \leq 1600 \text{ nm},$$

with particular emphasis on the region around 1550 nm. Within this wavelength range, the dispersive refractive indices of silicon and silicon dioxide are first evaluated and used consistently in the mode solver, supermode solver, and FDTD simulations. These platform assumptions define the waveguide cross-section, the optical confinement, and the expected fabrication constraints for the directional coupler. The main platform and design specifications used throughout the design flow are summarized in Table 1.

*Python Scripts and Dataset are available at: https://github.com/AmiEkh/Beam_Splitter_Project

Table 1: Target platform and design specifications for the silicon photonic beam-splitter design.

Parameter	**Target / Assumption**	**Purpose**
Material platform	SOI	Standard silicon photonics platform
Core material	Si	High-index guiding layer
Lower cladding	SiO_2	Buried oxide layer
Upper cladding	SiO_2	Oxide-cladded waveguide assumption
Silicon thickness	220 nm	Fixed device-layer thickness
Wavelength range	1500–1600 nm	Telecom-band operation
Central wavelength	1550 nm	Main design and optimization wavelength
Polarization	TE-like fundamental mode	Target guided mode
Waveguide type	Fully etched strip waveguide	Baseline waveguide geometry
Waveguide width	Selected by mode sweep	Chosen from n_{eff}, n_g, MFD, and field profile
Coupler type	Symmetric directional coupler	Four-port beam-splitter implementation
Coupler gap	Swept parameter	Controls coupling strength and beat length
Coupling length	Tuned parameter	Sets the 50/50 splitting condition
Target splitting ratio	50/50	Balanced beam-splitter operation
Access routing	S-bends	Connect compact coupler to fiber-array pitch
Fiber-array pitch	127 μm	Standard grating-coupler array pitch
Input/output couplers	Grating couplers	Vertical fiber probing
Layout tool	`gdsfactory`	Parametric hierarchical GDS generation
Simulation tools	Lumerical FDE/FDTD/EME	Mode analysis, supermode analysis, and full-wave verification
Test structures	Reference waveguides, bends, and couplers	Separate beam-splitter response from routing and coupling losses

The first design step is the analysis of a single silicon strip waveguide. For a fixed silicon thickness of 220 nm, the waveguide width is swept in order to study the supported guided modes. For each width, the effective index n_{eff}, group index n_g, mode-field diameters MFD_x and MFD_y, and electromagnetic field profiles are extracted. This step is used to choose a practical waveguide width that provides strong confinement, acceptable mode size, and robust fabrication tolerance. The selected waveguide width then becomes the baseline geometry for the directional coupler design.

After selecting the single-waveguide geometry, the directional coupler is investigated using even and odd supermode analysis. Two identical silicon waveguides are placed in parallel, and the gap g between them is swept. For each gap, the effective indices of the even and odd supermodes,

$$n_{\text{even}}(g), \qquad n_{\text{odd}}(g),$$

are computed. The supermode index splitting determines the coupling strength and the corresponding beat length. Using coupled-mode theory, the power-transfer length is estimated from the propagation-constant difference between the two supermodes. The approximate 50/50 coupling length is then obtained as a function of gap. This provides a theoretical starting point for selecting several candidate gaps and coupling lengths before full-vector electromagnetic verification.

The directional coupler is then converted into a complete four-port beam-splitter cell. In this layout, the coupling region is connected to input and output access waveguides using low-loss S-bends. The full geometry is generated parametrically using `gdsfactory`, allowing the coupler gap, coupling length, bend geometry, routing pitch, and labeling to be controlled automatically. The generated GDS is imported into Lumerical for electromagnetic simulation. This enables an automated loop in which the initial coupled-mode-theory values are used as starting points, and the coupling length is then tuned to approach the desired 50/50 transmission condition.

Once the coupling lengths are selected for the chosen set of gaps and waveguide widths, the final chip layout is assembled. Each beam splitter is placed as an independent hierarchical cell, and the optical ports are routed to standard grating-coupler access structures with a pitch of

$$127\ \mu\text{m}.$$

This makes the devices compatible with fiber-array probing. In addition to the main beam-splitter devices, dedicated test structures are included on the chip. These test structures are used to separate the performance of the beam splitter from other sources of loss, such as grating-coupler loss, waveguide propagation loss, S-bend loss, and routing imbalance. The function of each test structure and the corresponding measurement purpose are described in the chip-level layout section.

The complete workflow therefore connects theory, simulation, and layout in a single design pipeline. The platform definition and single-waveguide analysis determine the basic guided mode. The supermode analysis and coupled-mode theory provide the initial directional-coupler dimensions. Electromagnetic simulations refine and verify the coupling behavior. Finally, the optimized beam splitters are integrated into a hierarchical GDS layout with grating-coupler arrays, labels, and test structures for experimental characterization.

1 Waveguide Design

Material Model

The first step in the waveguide design is to define accurate material models for the silicon core and silicon dioxide cladding. In this work, the refractive-index data for both Si and SiO_2 are taken from the Palik material database and fitted inside Lumerical MODE. The fitting is performed over the wavelength range

$$1200\ \text{nm} \leq \lambda \leq 1700\ \text{nm},$$

which provides a sufficiently broad fitting window around the final operating band. The fitted material models are then used in the actual design simulations over the target wavelength range

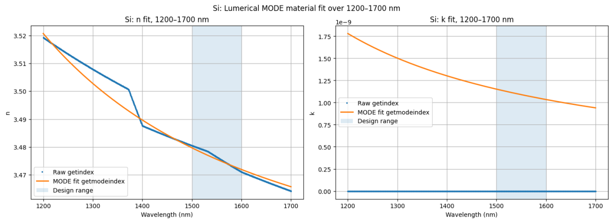
$$1500\ \text{nm} \leq \lambda \leq 1600\ \text{nm}.$$

Using a wider fitting range than the final simulation range helps avoid edge artifacts in the dispersive material model and gives a smooth representation of the refractive index across the wavelengths of interest. The silicon material is used as the high-index waveguide core, while SiO_2 is used as both the lower and upper cladding material. The complex refractive index is written as

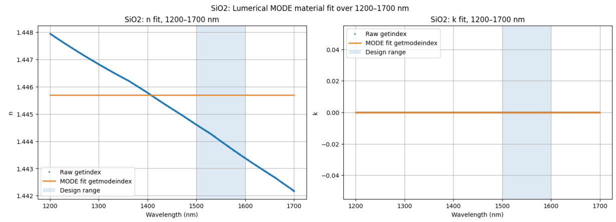
$$\tilde{n}(\lambda) = n(\lambda) + ik(\lambda),$$

where n is the real refractive index and k is the extinction coefficient.

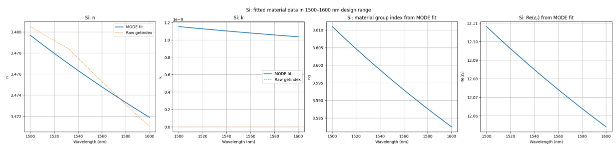
The Palik raw data and the fitted Lumerical MODE material models are shown in Fig. 1. The first row shows the material fits over the full fitting range from 1200 nm to 1700 nm for Si and



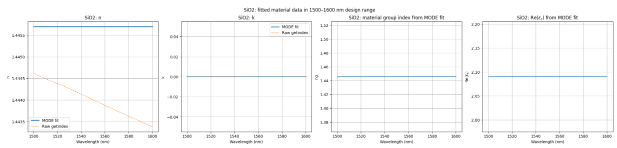
(a) Si fit over 1200–1700 nm.



(b) SiO_2 fit over 1200–1700 nm.



(c) Si properties over 1500–1600 nm.



(d) SiO_2 properties over 1500–1600 nm.

Figure 1: Summary of the Lumerical MODE material fits used in the waveguide simulations. The top row shows the Palik raw data together with the fitted material models over the full 1200–1700 nm fitting range, with the shaded region indicating the 1500–1600 nm design band. The bottom row shows the fitted material properties evaluated directly in the design band, including the real refractive index n, extinction coefficient k, material group index n_g, and real part of the relative permittivity $\mathrm{Re}(\varepsilon_r)$.

SiO_2. The shaded region indicates the final 1500–1600 nm design band. The second row shows the fitted material properties evaluated directly inside the design range. These plots include the fitted refractive index, extinction coefficient, material group index, and real part of the relative permittivity.

For silicon, the fitted real index decreases from approximately $n \approx 3.48$ near 1500 nm to $n \approx 3.47$ near 1600 nm. The fitted extinction coefficient is very small in this band, on the order of 10^{-9}, so material absorption is negligible for the present passive waveguide simulations. For SiO_2, the fitted refractive index is approximately $n \approx 1.445$ and the extinction coefficient is approximately zero in the design band.

The material group index is calculated from

$$n_g = n - \lambda \frac{dn}{d\lambda},$$

and the relative permittivity is calculated from

$$\varepsilon_r = \tilde{n}^2.$$

Since the fitted absorption is negligible in the selected wavelength range, the real part of the relative permittivity is approximately

$$\mathrm{Re}(\varepsilon_r) \approx n^2.$$

For silicon, the fitted material group index in the 1500–1600 nm range is approximately $n_g \approx$ 3.58–3.61, and $\mathrm{Re}(\varepsilon_r)$ decreases from approximately 12.11 to 12.05. For SiO_2, the fitted model is nearly dispersionless in the design band, giving an approximately constant group index and relative permittivity.

At this stage of the design, the oxide cladding is treated as effectively infinite. This approximation is useful for the initial mode analysis because it isolates the intrinsic waveguide behavior from finite-cladding and substrate effects. In the final simulation setup, the oxide thickness will be chosen large enough such that further increases in cladding thickness do not noticeably change the effective index, mode profile, mode-field diameter, or coupling behavior. Therefore, the infinite-cladding approximation is used only as an initial design assumption and will later be verified against a finite-cladding structure.

The initial material stack is therefore a silicon strip waveguide embedded in SiO_2, with a fixed silicon device-layer thickness of

$$t_{\mathrm{Si}} = 220 \text{ nm}.$$

The waveguide width is treated as a design parameter and is swept in the single-waveguide analysis.

Single-Waveguide Analysis

After defining the material models, the next step is to analyze a single silicon strip waveguide embedded in SiO_2. The silicon thickness is fixed to

$$t_{\mathrm{Si}} = 220 \text{ nm},$$

and the waveguide width is swept in order to study the modal properties of the structure. For each width, the eigenmodes are calculated using the finite-difference eigenmode solver. The main extracted quantities are the effective index n_{eff}, the group index n_g, the horizontal mode-field diameter MFD_x, and the vertical mode-field diameter MFD_y.

The width sweep was performed using the notebook `single_wg_mode_sweep.ipynb`, while the detailed electric-field and magnetic-field profiles, together with the MFD calculations and field-cut extraction, were obtained using the notebook `single_wg_mode.ipynb`.

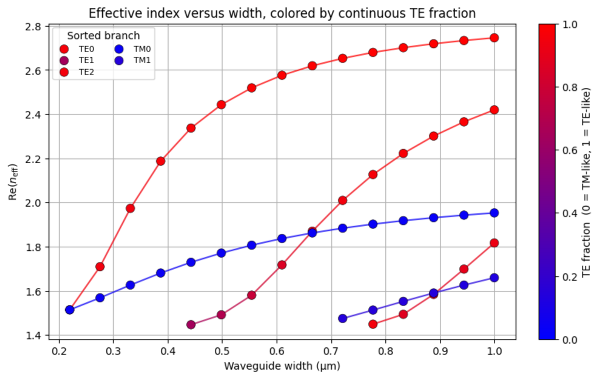
The calculated modes are sorted into continuous mode branches according to their effective index and polarization content. The transverse-electric fraction is used as an indicator of the polarization character of each mode, where a value close to 1 corresponds to a TE-like mode and a value close to 0 corresponds to a TM-like mode.

Figure 2 summarizes the main single-waveguide modal properties as a function of waveguide width. The real part of the effective index increases with width because the optical mode becomes more confined inside the high-index silicon core. At small widths, the effective index is close to the cladding index and the mode is weakly confined. As the width increases, the mode becomes more localized in the silicon region and n_{eff} approaches larger values.

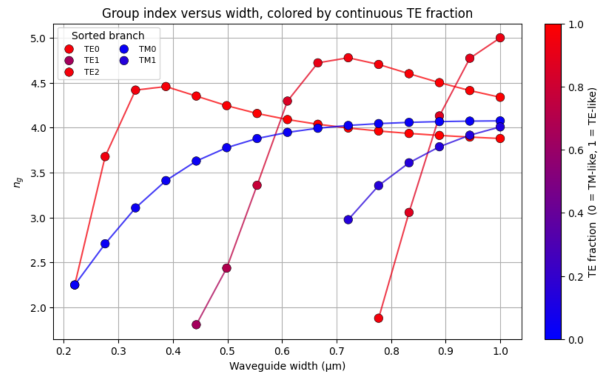
The group index also depends strongly on the waveguide width. The fundamental TE-like mode has a relatively large group index due to the combined effect of material dispersion and waveguide dispersion in the silicon strip waveguide. This parameter is important for later coupler design because it determines the propagation delay and the wavelength dependence of the optical phase.

The horizontal and vertical mode-field diameters show the confinement behavior more directly. For the fundamental TE-like mode, MFD_x decreases strongly as the waveguide width increases from the weakly confined regime to the more tightly confined regime. This indicates that wider waveguides provide stronger lateral confinement. The vertical mode-field diameter MFD_y is also reduced as the mode becomes more confined, although the vertical confinement is mainly controlled by the fixed 220 nm silicon thickness and the vertical index contrast with the oxide cladding.

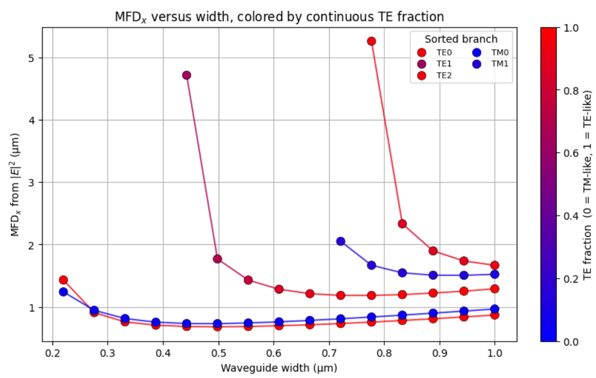
To evaluate the degree of lateral confinement more clearly, the ratio MFD_x/w is plotted for the fundamental TE0 mode, where w is the waveguide width. This result is shown in Figure 3. For narrow waveguides, MFD_x/w is much larger than one, meaning that a large part of the mode extends outside the physical waveguide width. As the width increases, this ratio approaches one, indicating that the optical mode becomes more confined laterally inside the silicon core.



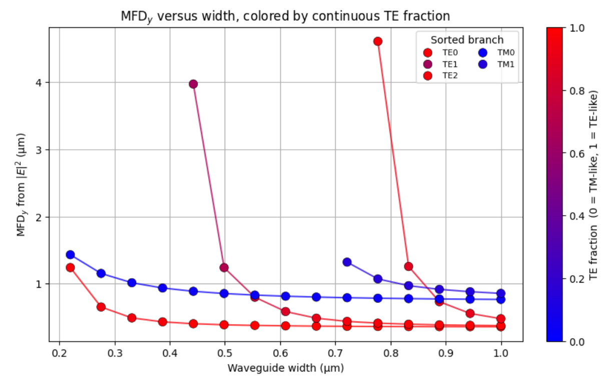
(a) Effective index n_{eff}.



(b) Group index n_g.



(c) Horizontal mode-field diameter MFD_x.



(d) Vertical mode-field diameter MFD_y.

Figure 2: Single-waveguide modal properties as a function of silicon waveguide width for a fixed silicon thickness of $t_{\mathrm{Si}} = 220$ nm. The four panels show the effective index, group index, horizontal mode-field diameter, and vertical mode-field diameter. The mode branches are colored by the continuous TE fraction, where red corresponds to TE-like modes and blue corresponds to TM-like modes.

The modal field profiles are also inspected for the two candidate widths,

$$w = 300 \text{ nm} \qquad \text{and} \qquad w = 500 \text{ nm}.$$

For each width, the real electric- and magnetic-field components of the fundamental mode are plotted, together with the normalized electric-field intensity $|E|^2$ and its horizontal and vertical cuts. These field plots are useful because they show the spatial confinement directly, rather than only through the scalar quantities n_{eff}, n_g, and MFD.

The four field and cut plots are grouped in Fig. 4. The top row corresponds to the 300 nm-wide waveguide, and the bottom row corresponds to the 500 nm-wide waveguide. For each width, the left panel shows the real electric- and magnetic-field components of the fundamental mode, while the right panel shows the normalized electric-field intensity $|E|^2$ and the corresponding horizontal and vertical cuts.

For the 300 nm-wide waveguide, the mode is more weakly confined laterally, and the optical field extends farther into the surrounding $\mathrm{SiO_2}$ cladding. This larger evanescent tail is beneficial for directional coupling because it allows two adjacent waveguides to interact at a larger physical gap. In contrast, for the 500 nm-wide waveguide, the mode is more strongly confined inside the silicon

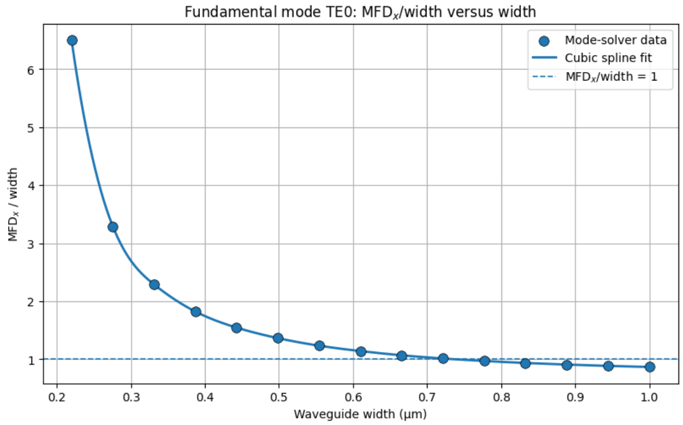


Figure 3: Ratio MFD_x/w versus waveguide width for the fundamental TE0 mode. The dashed horizontal line indicates $\mathrm{MFD}_x/w = 1$.

core. The extracted mode-field diameter is smaller, and the field decays more rapidly into the oxide cladding. This stronger confinement is beneficial for compact bends and reduces sensitivity to small width perturbations, but it also means that smaller waveguide gaps are required in the double-waveguide coupler to obtain sufficient coupling.

Based on these results, two important design regimes can be identified. A waveguide width near

$$w = 500 \text{ nm}$$

provides stronger confinement of the fundamental TE0 mode. This is beneficial because the mode is better localized inside the silicon core, which gives better compatibility with compact bends and reduces the sensitivity of the effective index to small width variations. However, the stronger confinement also means that the evanescent tail decays more rapidly in the oxide cladding. Therefore, in a directional coupler or double-waveguide geometry, a smaller gap is required to obtain sufficiently strong coupling.

In contrast, a waveguide width near

$$w = 300 \text{ nm}$$

has a larger horizontal mode-field diameter and a larger evanescent tail. This can be advantageous for coupling because the same coupling strength can be achieved with a larger physical gap. A larger gap is generally easier to fabricate and can be less sensitive to small gap variations. However, the narrower waveguide is more weakly confined, more sensitive to width variations, and generally less favorable for compact low-loss bends.

The sensitivity of the effective index to waveguide-width variations is shown in Figure 5. Around $w = 300$ nm, a width perturbation of ± 5 nm produces a larger change in n_{eff} than around $w = 500$ nm. From the spline fit, the extracted values are approximately

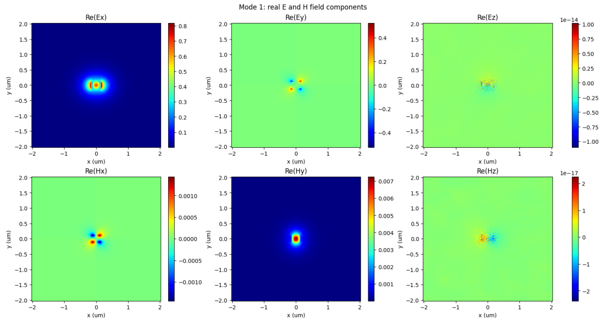
$$n_{\text{eff}}(300 \text{ nm}) \approx 1.8216, \qquad \Delta n_{\text{eff}}(\pm 5 \text{ nm}) \approx 4.87 \times 10^{-2},$$

and

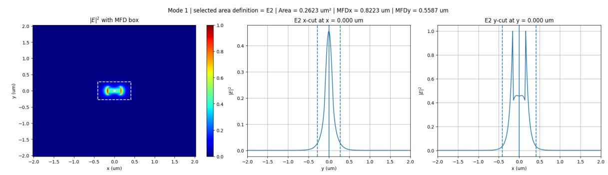
$$n_{\text{eff}}(500 \text{ nm}) \approx 2.44498, \qquad \Delta n_{\text{eff}}(\pm 5 \text{ nm}) \approx 1.58 \times 10^{-2}.$$

Equivalently, the local sensitivities are approximately

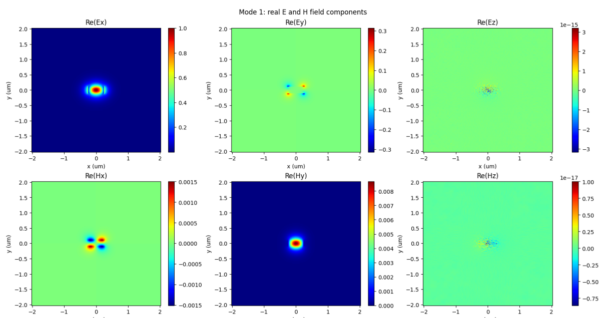
$$\left.\frac{dn_{\text{eff}}}{dw}\right|_{w=300 \text{ nm}} \approx 4.87 \times 10^{-3} \text{ nm}^{-1},$$



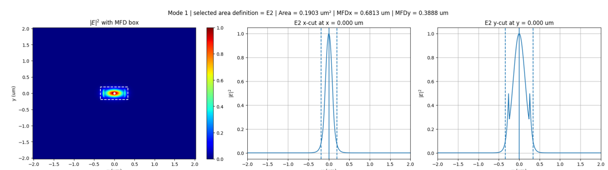
(a) 300 nm waveguide: real E and H field components.



(b) 300 nm waveguide: normalized $|E|^2$ and x/y cuts.



(c) 500 nm waveguide: real E and H field components.



(d) 500 nm waveguide: normalized $|E|^2$ and x/y cuts.

Figure 4: Fundamental-mode field profiles and intensity cuts for the two candidate waveguide widths. The top row shows the 300 nm-wide waveguide and the bottom row shows the 500 nm-wide waveguide. The left column shows the real electric- and magnetic-field components, while the right column shows the normalized electric-field intensity $|E|^2$ together with the horizontal and vertical cuts used to extract the mode-field diameter.

and

$$\left.\frac{dn_{\mathrm{eff}}}{dw}\right|_{w=500\ \mathrm{nm}} \approx 1.58 \times 10^{-3}\ \mathrm{nm}^{-1}.$$

Therefore, the 500 nm waveguide is expected to be more tolerant to fabrication-induced width variations.

The main trade-off is therefore between confinement and coupling strength. A 500 nm waveguide provides better confinement, lower sensitivity to width errors, and better compatibility with compact bends, but it requires smaller coupling gaps because of the shorter evanescent tail. A 300 nm waveguide provides a larger evanescent field and can enable coupling at larger gaps, but it is less confined and more sensitive to fabrication variations.

For the present design, the 500 nm width is selected as the main candidate for the fundamental waveguide. Although this width supports additional higher-order modes, the fundamental TE0 mode remains the primary design mode. The higher-order modes have different polarization content and weaker overlap with the input fundamental mode, so their excitation is expected to be limited if the input/output transitions and bends are designed adiabatically. This choice will be further checked in the double-waveguide and full-device simulations.

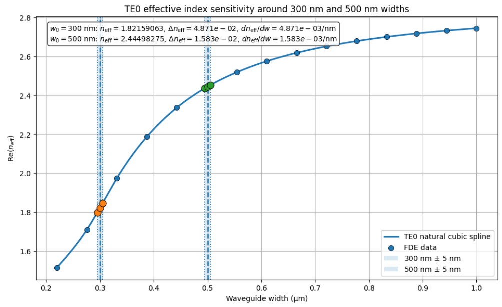


Figure 5: Sensitivity of the TE0 effective index to waveguide-width variations around 300 nm and 500 nm. The shaded regions indicate ±5 nm width variations around each nominal width.

Double-Waveguide Analysis

After selecting candidate single-waveguide widths, the next step is to analyze two identical parallel silicon waveguides separated by a gap. This structure forms the directional-coupler section of the beam splitter. In the coupled-waveguide region, optical power is transferred between the two waveguides through the overlap of their evanescent fields in the SiO_2 cladding.

The physics of the directional coupler can be described using coupled-mode theory. If the two isolated waveguides are identical and phase matched, their uncoupled propagation constants are equal,

$$\beta_1 = \beta_2 = \beta_0.$$

The slowly varying modal amplitudes $a_1(z)$ and $a_2(z)$ satisfy

$$\frac{da_1}{dz} = -i\kappa a_2, \qquad \frac{da_2}{dz} = -i\kappa a_1,$$

where κ is the coupling coefficient. For excitation in waveguide 1 at $z = 0$, the solution is

$$a_1(z) = \cos(\kappa z), \qquad a_2(z) = -i\sin(\kappa z).$$

Therefore, the powers in the two waveguides are

$$P_1(z) = \cos^2(\kappa z), \qquad P_2(z) = \sin^2(\kappa z).$$

A 50/50 splitter is obtained when

$$P_1 = P_2 = \frac{1}{2},$$

which gives the half-power coupling length

$$L_{50/50} = \frac{\pi}{4\kappa}.$$

The same coupling coefficient can also be extracted from the even and odd supermodes of the coupled double-waveguide system. For two identical phase-matched waveguides, the symmetric and antisymmetric supermodes have propagation constants

$$\beta_{\text{even}} = \beta_0 + \kappa, \qquad \beta_{\text{odd}} = \beta_0 - \kappa.$$

The propagation-constant splitting is therefore

$$\Delta\beta = \beta_{\text{even}} - \beta_{\text{odd}} = 2\kappa.$$

Using

$$\beta = k_0 n_{\text{eff}} = \frac{2\pi}{\lambda_0} n_{\text{eff}},$$

the coupling coefficient is

$$\kappa = \frac{\Delta\beta}{2} = \frac{\pi}{\lambda_0}\left(n_{\text{eff,even}} - n_{\text{eff,odd}}\right).$$

Equivalently, the 50/50 coupling length can be calculated directly from the supermode index splitting:

$$L_{50/50} = \frac{\pi}{2\Delta\beta} = \frac{\lambda_0}{4\Delta n_{\text{eff}}},$$

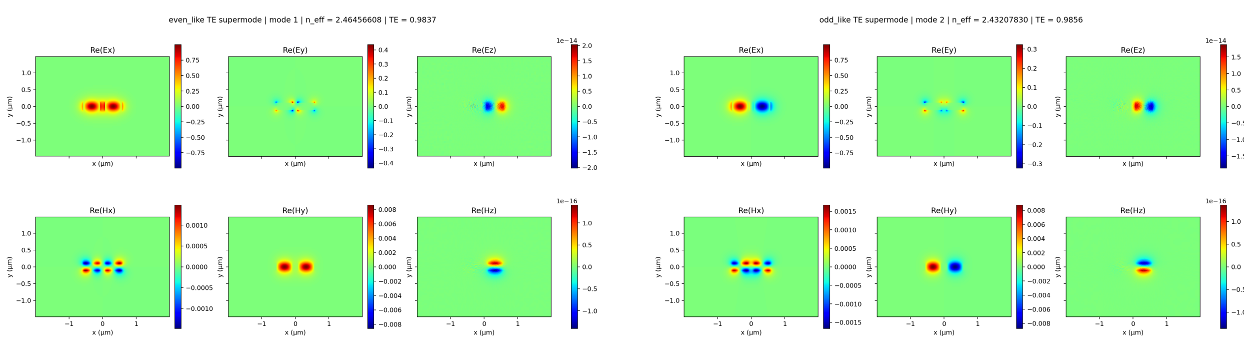
where

$$\Delta n_{\text{eff}} = n_{\text{eff,even}} - n_{\text{eff,odd}}.$$

This supermode method is used here to estimate the required coupling length as a function of gap.

The gap sweep was performed using the notebook `double_wg_mode_sweep.ipynb`, while the detailed electric-field and magnetic-field profiles, and field-cut extraction, were obtained using the notebook `double_wg_mode.ipynb`.

For waveguide width of 500 nm a fixed gap of $g = 150$ nm, the even-like and odd-like TE supermodes are first calculated to verify the coupling behavior. Figure 6 shows the real electric- and magnetic-field components of the even-like and odd-like TE supermodes in a two-column tile format. The even-like mode has the dominant transverse electric field with the same sign in the two waveguides, while the odd-like mode has the dominant transverse electric field with opposite signs in the two waveguides. This symmetry difference produces the effective-index splitting between the two supermodes.

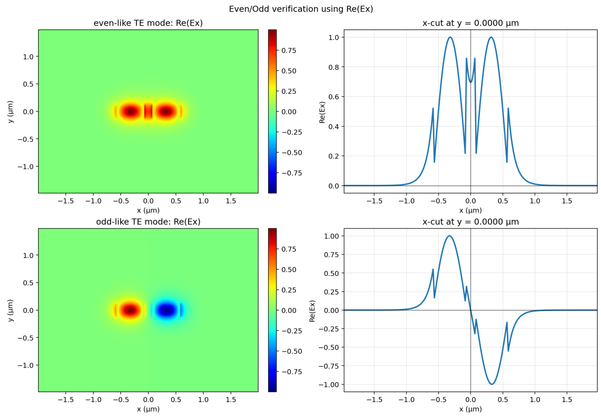


(a) Even-like TE supermode.

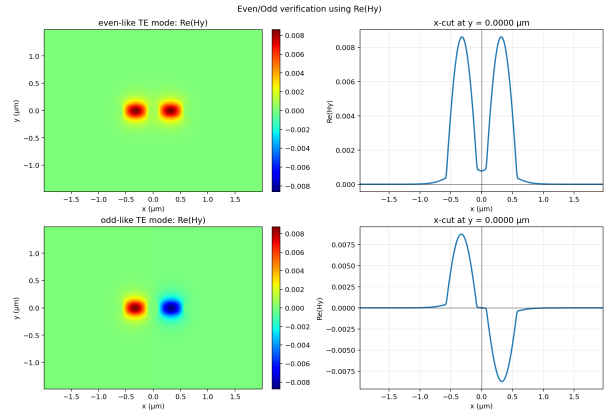
(b) Odd-like TE supermode.

Figure 6: Real electric- and magnetic-field components of the even-like and odd-like TE supermodes for the double silicon waveguide with a fixed gap of $g = 150$ nm. The two panels are arranged in a 1×2 tile format.

The even/odd symmetry is further verified by plotting the dominant field components along the horizontal cut through the center of the waveguides. Figure 7 shows the verification using $\mathrm{Re}(E_x)$ and $\mathrm{Re}(H_y)$ in a two-column tile format. For the even-like supermode, $\mathrm{Re}(E_x)$ has the same sign in the two waveguides. For the odd-like supermode, $\mathrm{Re}(E_x)$ changes sign between the two waveguides. The $\mathrm{Re}(H_y)$ cut gives the corresponding magnetic-field symmetry verification.



(a) Even/odd supermode verification using $\mathrm{Re}(E_x)$.



(b) Even/odd supermode verification using $\mathrm{Re}(H_y)$.

Figure 7: Verification of the even/odd supermode symmetry using horizontal cuts of $\mathrm{Re}(E_x)$ and $\mathrm{Re}(H_y)$. The two panels are arranged in a 1×2 tile format.

The gap is then swept for the two candidate waveguide widths,

$$w = 300 \text{ nm} \qquad \text{and} \qquad w = 500 \text{ nm}.$$

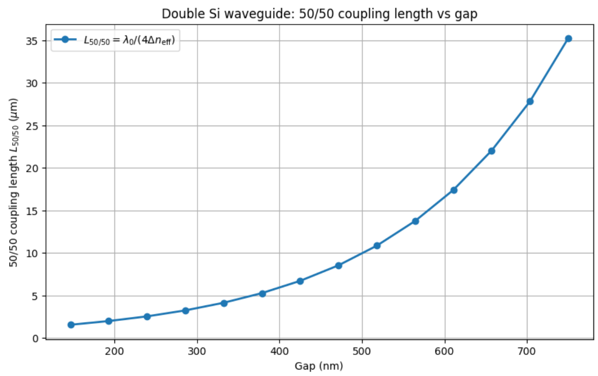
For each width, the even and odd supermode indices are extracted, and the coupling length is calculated from

$$L_{50/50} = \frac{\lambda_0}{4\Delta n_{\mathrm{eff}}}.$$

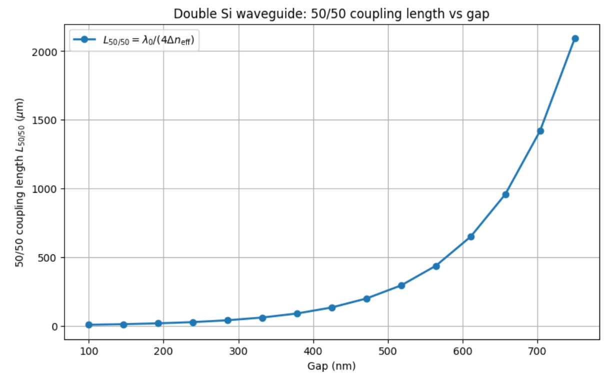
The corresponding coupling coefficient is calculated from

$$\kappa = \frac{\pi}{\lambda_0}\Delta n_{\mathrm{eff}}.$$

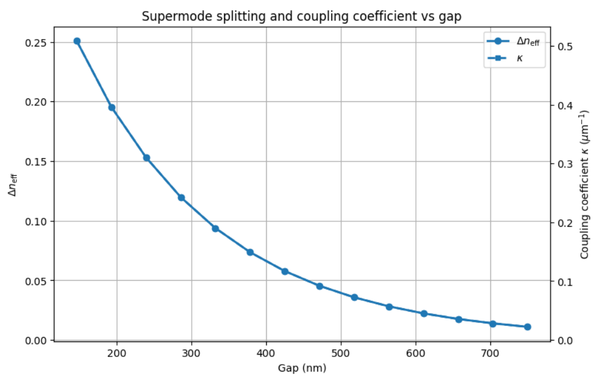
Figure 8 summarizes the gap-sweep results for both waveguide widths. The first row shows the 50/50 coupling length versus gap, while the second row shows the corresponding supermode splitting and coupling coefficient. The 300 nm waveguide has a larger evanescent tail, so it can achieve strong coupling at relatively larger gaps. The 500 nm waveguide is more strongly confined, so its evanescent field decays faster in the oxide and smaller gaps are required to obtain the same coupling strength.



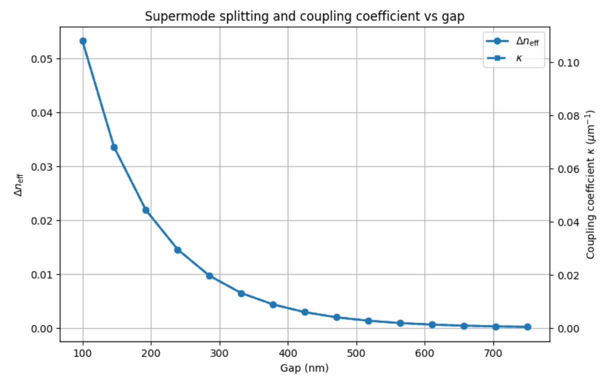
(a) $L_{50/50}$ versus gap for $w = 300$ nm.



(b) $L_{50/50}$ versus gap for $w = 500$ nm.



(c) $\Delta n_{\rm eff}$ and κ versus gap for $w = 300$ nm.



(d) $\Delta n_{\rm eff}$ and κ versus gap for $w = 500$ nm.

Figure 8: Gap-sweep results for the double silicon waveguide. The top row shows the 50/50 coupling length $L_{50/50}$ versus gap for the 300 nm and 500 nm waveguide widths. The bottom row shows the corresponding supermode effective-index splitting $\Delta n_{\rm eff}$ and coupling coefficient κ.

Fabrication errors can affect the coupler in two main ways. First, width variations change the isolated-waveguide effective index. If the two waveguides are no longer identical, a propagation-constant mismatch is introduced:

$$\Delta\beta_{\rm mismatch} = \beta_1 - \beta_2.$$

In the presence of mismatch, the coupled-mode equations become

$$\frac{da_1}{dz} = -i\frac{\Delta\beta_{\rm mismatch}}{2}a_1 - i\kappa a_2,$$

$$\frac{da_2}{dz} = +i\frac{\Delta\beta_{\rm mismatch}}{2}a_2 - i\kappa a_1.$$

The maximum transferable power is then reduced to approximately

$$P_{2,\max} = \frac{\kappa^2}{\kappa^2 + (\Delta\beta_{\rm mismatch}/2)^2}.$$

Therefore, robust coupling requires

$$\kappa \gg \frac{|\Delta\beta_{\rm mismatch}|}{2}.$$

This means that stronger coupling is more tolerant to width-induced phase mismatch.

Second, gap variations directly change the coupling coefficient κ. Since κ depends strongly on the evanescent overlap, large gaps produce weak coupling and can make the coupling length very sensitive to small gap errors. Smaller gaps give stronger coupling, shorter coupling lengths, and better tolerance to phase mismatch. However, very small gaps can be more challenging to fabricate and may increase sensitivity to absolute gap errors. Therefore, the selected gap should be small enough to provide strong coupling, but not so small that fabrication becomes unreliable.

To quantify this robustness, the coupling coefficient is compared with the estimated phase-mismatch scale caused by width variations. The width-mismatch detuning is estimated as

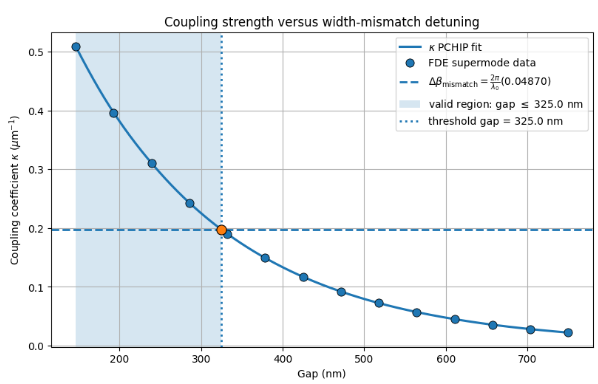
$$\Delta\beta_{\text{mismatch}} = \frac{2\pi}{\lambda_0}\Delta n_{\text{eff,mismatch}}.$$

The safe region is defined as the gap range where

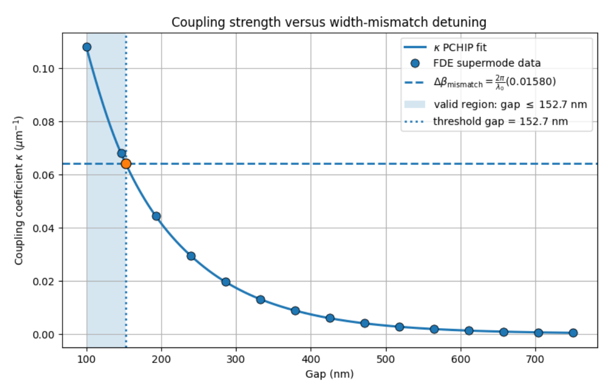
$$\kappa > \frac{|\Delta\beta_{\text{mismatch}}|}{2}.$$

In this regime, the coupling strength dominates over the width-induced phase mismatch, so the directional coupler is expected to be more robust against fabrication tolerance.

Figure 9 shows the safe-zone analysis for both waveguide widths. The shaded regions indicate the gap ranges where the coupling coefficient is larger than the estimated width-mismatch detuning. The 300 nm waveguide can remain robust over a larger gap range because of its stronger evanescent overlap. The 500 nm waveguide requires a smaller gap to reach the same robustness condition, but it benefits from better single-waveguide confinement and lower sensitivity of the isolated mode to width variations.



(a) Safe-zone analysis for $w = 300$ nm.



(b) Safe-zone analysis for $w = 500$ nm.

Figure 9: Robustness analysis against width-mismatch detuning for the 300 nm and 500 nm waveguide widths. The shaded region indicates the gap range where the coupling coefficient κ is larger than the estimated phase-mismatch scale, giving a more robust directional coupler.

The numerical design candidates extracted from the gap sweep are summarized directly in Tables 2a and 2b. The tables are placed inside the text using the `[H]` placement option to prevent them from floating to the end of the document. For the 300 nm waveguide, larger gaps can still provide short coupling lengths because the mode is less confined and has stronger evanescent overlap. For the 500 nm waveguide, smaller gaps are required to obtain compact couplers.

(a) Candidate 50/50 coupling lengths for the w = 300 nm silicon waveguide.

Waveguide width	**Gap**	$L_{50/50}$
300 nm	150 nm	1.574 μm
300 nm	200 nm	2.059 μm
300 nm	250 nm	2.677 μm

(b) Candidate 50/50 coupling lengths for the w = 500 nm silicon waveguide.

Waveguide width	**Gap**	$L_{50/50}$
500 nm	125 nm	8.911 μm
500 nm	150 nm	11.960 μm
500 nm	200 nm	18.895 μm

Table 2: Candidate 50/50 coupling lengths for the two selected waveguide widths. The highlighted entries indicate the preferred design points for each width.

Overall, the double-waveguide analysis confirms the main trade-off observed in the single-waveguide study. Narrower waveguides provide larger evanescent tails and allow coupling at larger gaps, while wider waveguides provide stronger confinement and better compatibility with compact routing and bends, but require smaller gaps to obtain strong coupling. From the robustness analysis, the selected gap should lie inside the safe zone where the coupling coefficient is larger than the width-mismatch detuning. Therefore, for the 300 nm waveguide, gaps around 150–250 nm are compact and robust candidates, while for the 500 nm waveguide, gaps around 125–200 nm should be evaluated further using full directional-coupler simulations. Based on the highlighted design points in Table 2, the selected coupling gaps for the next-stage beam-splitter simulations are $g = 250$ nm for the 300 nm waveguide and $g = 150$ nm for the 500 nm waveguide.

2 Beam-Splitter Layout Generation and Simulation Workflow

Layout Generation

After selecting the waveguide widths, gap candidates, and initial coupling lengths from the coupled-mode analysis, the next step is to build the full beam-splitter geometry and prepare it for numerical tuning. The beam splitter is designed as a symmetric directional coupler, where the two input and two output waveguides are connected to the central coupling region through identical S-bends. This symmetric layout keeps the two optical paths geometrically equivalent and helps preserve a balanced splitting response between the two output ports.

The outer waveguide pitch is chosen such that the interaction between the uncoupled input and output arms is negligible before and after the directional-coupler region. In this design, the pitch is set to the lowest practical value of

$$P = 4\ \mu\text{m},$$

which keeps the layout compact while still making the residual coupling between the separated arms negligible. Reducing the pitch also reduces the lateral displacement required in the S-bends, and therefore reduces the longitudinal routing length. However, the local bend radius must remain sufficiently large to avoid radiation loss. As a first design estimate, the S-bends are chosen such that the minimum bend radius satisfies

$$R_{\min} > 50\ \mu\text{m}.$$

The required S-bend length can be estimated from this bend-radius constraint. For a compact first design, each S-bend is described by a smooth raised-cosine trajectory,

$$y(x) = \frac{\Delta y}{2}\left[1 - \cos\left(\frac{\pi x}{L_{\text{SB}}}\right)\right], \qquad 0 \le x \le L_{\text{SB}},$$

where L_{SB} is the longitudinal S-bend length and Δy is the lateral displacement of one waveguide arm. The local radius of curvature is

$$R(x) = \frac{\left[1 + (y'(x))^2\right]^{3/2}}{|y''(x)|}.$$

For moderate slopes, the slope correction in the numerator can be neglected for a first estimate. Since

$$y''(x) = \frac{\Delta y}{2}\left(\frac{\pi}{L_{\mathrm{SB}}}\right)^2 \cos\left(\frac{\pi x}{L_{\mathrm{SB}}}\right),$$

the maximum curvature occurs near the ends of the S-bend, giving

$$R_{\min} \approx \frac{2L_{\mathrm{SB}}^2}{\pi^2 \Delta y}.$$

Therefore, for a target minimum bend radius $R_{\min}$, the required S-bend length is estimated as

$$L_{\mathrm{SB}} \gtrsim \sqrt{\frac{\pi^2 R_{\min} \Delta y}{2}}.$$

In the symmetric beam-splitter layout, the outer port pitch is P, while the center-to-center separation in the coupling region is

$$s = w + g,$$

where w is the waveguide width and g is the physical gap between the two waveguides. Therefore, each waveguide arm must move laterally by

$$\Delta y = \frac{P - s}{2} = \frac{P - (w + g)}{2}.$$

Using the chosen pitch $P = 4\ \mu$m and requiring $R_{\min} \geq 50\ \mu$m, the S-bend length can be estimated for each design.

For example, for the design

$$w = 0.300\ \mu\text{m}, \qquad g = 0.250\ \mu\text{m},$$

the center-to-center separation in the coupling region is

$$s = w + g = 0.550\ \mu\text{m}.$$

Thus, the lateral displacement of each arm is

$$\Delta y = \frac{4.000 - 0.550}{2} = 1.725\ \mu\text{m}.$$

The required S-bend length is then approximately

$$L_{\mathrm{SB}} \gtrsim \sqrt{\frac{\pi^2 (50)(1.725)}{2}} = 20.6\ \mu\text{m}.$$

Therefore, an S-bend length of about

$$L_{\mathrm{SB}} \approx 21\ \mu\text{m}$$

is the minimum first estimate for this geometry. In practice, a slightly longer value, such as

$$L_{\mathrm{SB}} = 25\ \mu\mathrm{m},$$

can be used to provide margin against radiation loss and discretization effects in the layout and simulation.

The complete beam-splitter layout consists of three main regions: an input S-bend section, a straight coupling section, and an output S-bend section. The straight coupling section has length L_c, while the physical gap between the two waveguides is g. For each selected gap, the initial value of L_c is taken from the coupled-mode-theory estimate obtained from the even–odd supermode splitting.

However, the CMT value of L_c is only an initial guess for the complete routed beam splitter. In the ideal straight directional-coupler calculation, the two waveguides are assumed to be parallel and separated by a fixed gap over the full interaction length. In the actual layout, the waveguides approach the coupling region gradually through the S-bends. Therefore, the separation is not initialized abruptly, and part of the optical power transfer can already begin during the transition into the coupling region. As a result, the optimal coupling length of the full routed beam splitter is expected to be slightly shorter than the coupling length predicted by the ideal straight-coupler CMT model.

The layout generation is performed using the notebook

`BeamSplitter_layout.ipynb`.

This notebook generates the symmetric beam-splitter geometry for user-defined values of the waveguide width w, gap g, coupling length L_c, and S-bend longitudinal length L_{SB}. In addition to the full beam splitter, the notebook also generates a single-arm reference structure. This reference arm is useful for separating the excess loss of the full beam splitter from the propagation and bend losses associated with a single routed waveguide arm.

The generated GDS layout uses separate layers for the optical waveguide geometry, text labels, and optional port or alignment markers. In this implementation, the silicon waveguide pattern is written on the main device layer

$$\text{Si waveguide}: \qquad (1, 0),$$

where the first number is the GDS layer number and the second number is the datatype. The text labels used for identifying the device parameters and port names are written on

$$\text{Text labels}: \qquad (10, 0),$$

and optional port-marker or simulation-marker polygons can be placed on

$$\text{Markers}: \qquad (100, 0).$$

These layers are used only to organize the mask layout and the simulation import. In the Lumerical simulation step, the silicon device layer $(1, 0)$ is imported as the 220 nm-thick Si core, while the oxide cladding and substrate are defined directly inside the simulation environment.

As an example layout, the beam splitter is generated for

$$w = 0.300\ \mu\mathrm{m}, \qquad g = 0.250\ \mu\mathrm{m}, \qquad L_c = 2.67\ \mu\mathrm{m},$$

with the S-bend longitudinal length chosen from the bend-radius constraint. The corresponding layout is shown in Fig. 10. The four optical ports are labeled as the upper and lower input ports on the left side, and the upper and lower output ports on the right side. This geometry is used as an initial routed design for numerical simulation and coupling-length tuning.

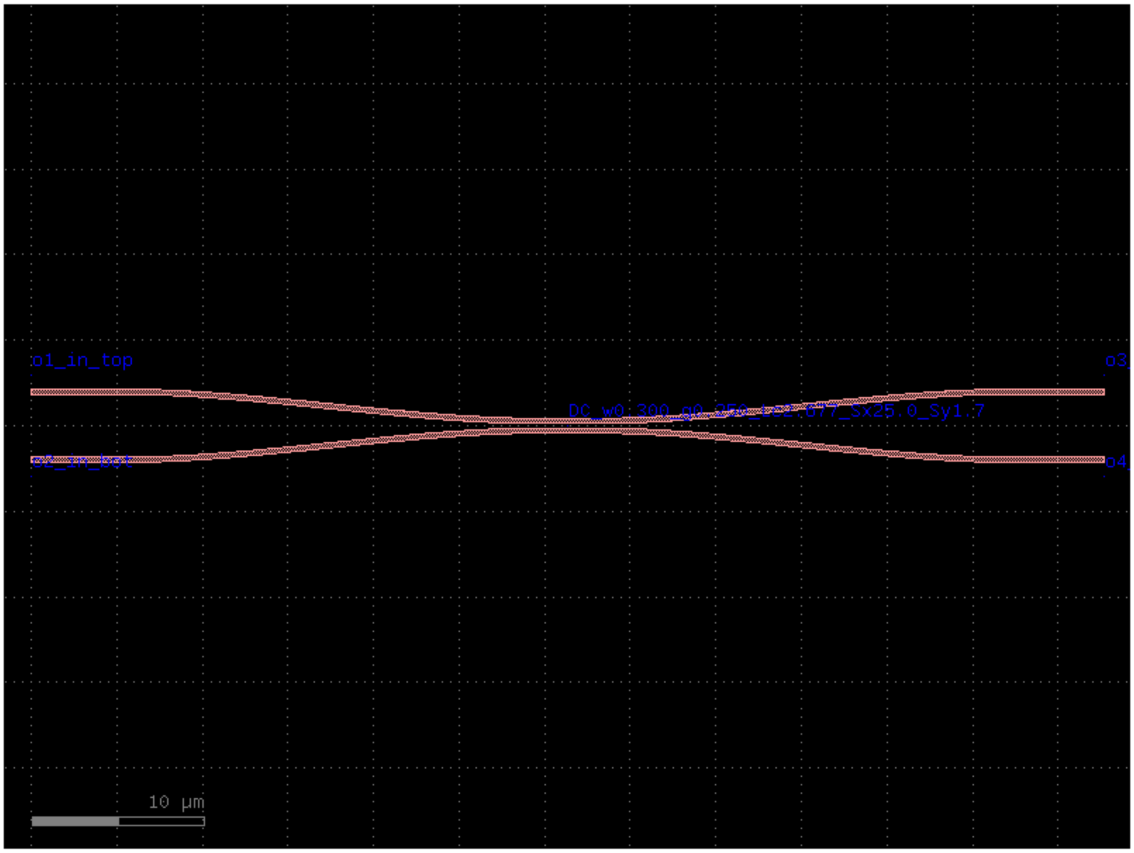


Figure 10: Example generated GDS layout of the symmetric directional-coupler beam splitter for $w = 0.300$ μm, $g = 0.250$ μm, and $L_c = 2.67$ μm. The input and output waveguides are connected to the central coupling region using identical S-bends, with an outer port pitch of $P = 4$ μm. The silicon waveguide pattern is placed on GDS layer $(1, 0)$, while text labels are placed on layer $(10, 0)$.

Simulation and Validation

After the GDS layout is generated, the beam splitter is imported into Lumerical for electromagnetic simulation. Two numerical approaches are considered: finite-difference time-domain simulation and eigenmode expansion simulation. The FDTD method provides a direct full-wave verification of the device response, including radiation loss, reflections, finite-cladding effects, and possible parasitic scattering. However, it can be computationally expensive for long directional-coupler structures with multiple ports. Therefore, for the initial tuning of the coupling length, the eigenmode expansion method is used as the more efficient approach.

The FDTD simulation workflow is automated using the notebooks

`FDTD_tuned_beam_splitter_infinite_cladding.ipynb`

and

`FDTD_tuned_beam_splitter_finite_cladding.ipynb`.

These notebooks import the generated GDS file into Lumerical, define the silicon and silicon-dioxide material models, set the waveguide geometry, add sources and monitors, and automate the four-port beam-splitter simulation. The infinite-cladding version is used for the first design tuning step, while the finite-cladding version can later be used to check the effect of a more realistic oxide stack.

For faster tuning, the EME simulation workflow is implemented using the notebooks

`EME_tuned_beam_splitter_infinite_cladding.ipynb`

and

`EME_tuned_beam_splitter_finite_cladding.ipynb`.

The EME method is especially suitable for this structure because the device is mainly composed of slowly varying waveguide sections and a straight directional-coupler region. This makes it possible to sweep the coupling length L_c and wavelength much more efficiently than with FDTD.

In the EME model, the imported GDS geometry is first extruded into a three-dimensional silicon waveguide structure with the correct device-layer thickness. The silicon core and silicon-dioxide cladding material models are then assigned, and the EME simulation region is placed around the complete routed beam splitter. The transverse simulation span is chosen large enough to include all four waveguide ports and the surrounding cladding. The mesh is refined around the silicon waveguides and the coupling region so that the narrow gap and the waveguide sidewalls are resolved accurately.

The four EME ports are placed at the straight input and output waveguide sections, after the S-bend routing. Each port solves the local guided mode at that cross section, and the EME solver propagates the modal amplitudes through the device. The central directional-coupler region is then swept in length to tune the power exchange between the two output waveguides. A representative EME setup, showing the imported geometry, mesh, and port locations, is shown in Fig. 11.

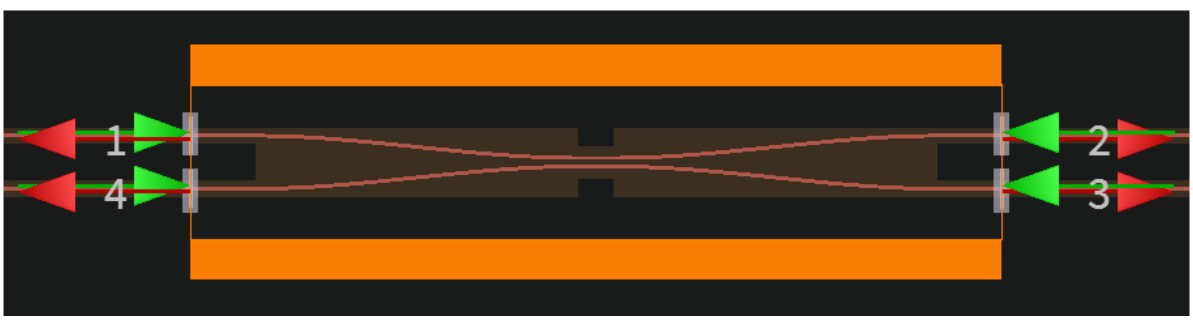


Figure 11: EME simulation setup for the four-port beam splitter. The imported GDS geometry is meshed, and the four EME ports are placed on the straight waveguide sections at the input and output sides of the routed directional coupler.

The S-matrix index ordering used by Lumerical must be mapped carefully to the physical port labels. In this simulation, the source-mode index mapping is

S-matrix index	Physical source mode
1	port 1, mode 1
2	port 4, mode 1
3	port 2, mode 1
4	port 3, mode 1

as shown in Fig. 12. Therefore, when the input is launched from physical port 1, the transmission to physical port 2 is extracted from

$$|s_{31}|^2,$$

and the transmission to physical port 3 is extracted from

$$|s_{41}|^2.$$

These two quantities are the through and cross output powers used for the beam-splitter tuning.

S-matrix index mapping

S-matrix index	Source mode	
1	port 1	mode 1
2	port 4	mode 1
3	port 2	mode 1
4	port 3	mode 1

Figure 12: S-matrix index mapping used in the EME simulation. The physical port order is not the same as the numerical S-matrix order, so the transmission terms must be selected using this mapping.

In this project, the initial tuning is performed using the infinite-cladding EME model. The objective of the tuning is to find the coupling length L_c that gives a balanced 50/50 output splitting ratio. The tuning starts from the coupled-mode-theory estimate of L_c for both selected waveguide widths,

$$w = 300 \text{ nm} \qquad \text{and} \qquad w = 500 \text{ nm}.$$

For the $w = 300$ nm design, the selected gap is

$$g = 250 \text{ nm},$$

and for the $w = 500$ nm design, the selected gap is

$$g = 150 \text{ nm}.$$

For each design case, the coupling length is swept around the initial coupled-mode-theory estimate, and the wavelength response is evaluated over

$$1500 \text{ nm} \leq \lambda \leq 1600 \text{ nm}.$$

The normalized through and cross splitting ratios are defined as

$$\eta_{\text{through}} = \frac{P_{\text{through}}}{P_{\text{through}} + P_{\text{cross}}}, \qquad \eta_{\text{cross}} = \frac{P_{\text{cross}}}{P_{\text{through}} + P_{\text{cross}}},$$

where P_{through} and P_{cross} are the transmitted powers at the two output ports. For excitation from physical port 1, these are extracted from the EME S-matrix as

$$P_{\text{through}} = |s_{31}|^2, \qquad P_{\text{cross}} = |s_{41}|^2,$$

using the S-matrix mapping described above. The ideal 50/50 beam-splitting condition is therefore

$$\eta_{\text{through}} = \eta_{\text{cross}} = 0.5.$$

Equivalently, the tuning can be performed by minimizing the splitting imbalance

$$\Delta_{\text{BS}} = |\eta_{\text{cross}} - \eta_{\text{through}}| \, .$$

Figure 13 shows the EME coupling-length sweeps for the two selected designs. For the $w = 300$ nm, $g = 250$ nm design, balanced 50/50 operation occurs at approximately

$$L_c = 1.77\ \mu\text{m}, \qquad 7.195\ \mu\text{m}, \qquad 12.617\ \mu\text{m}.$$

The spacing between consecutive 50/50 points gives an effective coupling period of approximately

$$L_\pi \approx 5.42\ \mu\text{m}.$$

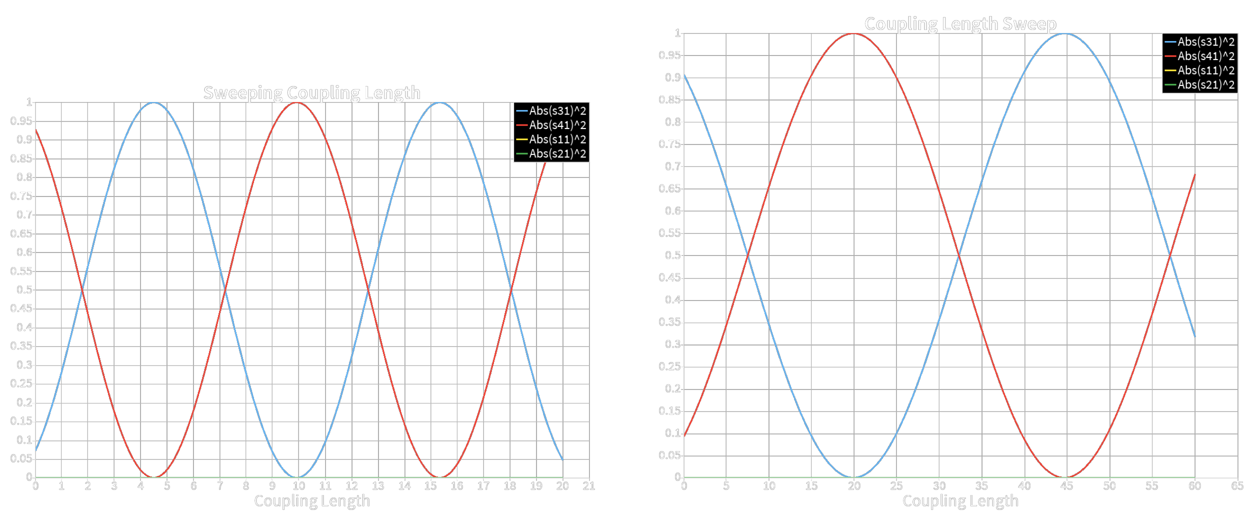
For the $w = 500$ nm, $g = 150$ nm design, the 50/50 points occur at approximately

$$L_c = 7.49\ \mu\text{m}, \qquad 32.26\ \mu\text{m}, \qquad 57.04\ \mu\text{m}.$$

The corresponding effective coupling period is

$$L_\pi \approx 24.77\ \mu\text{m}.$$

The much longer coupling period for the 500 nm waveguide is expected because the wider waveguide confines the optical mode more strongly, reducing the evanescent overlap between the two coupled waveguides.



(a) $w = 300$ nm, $g = 250$ nm.

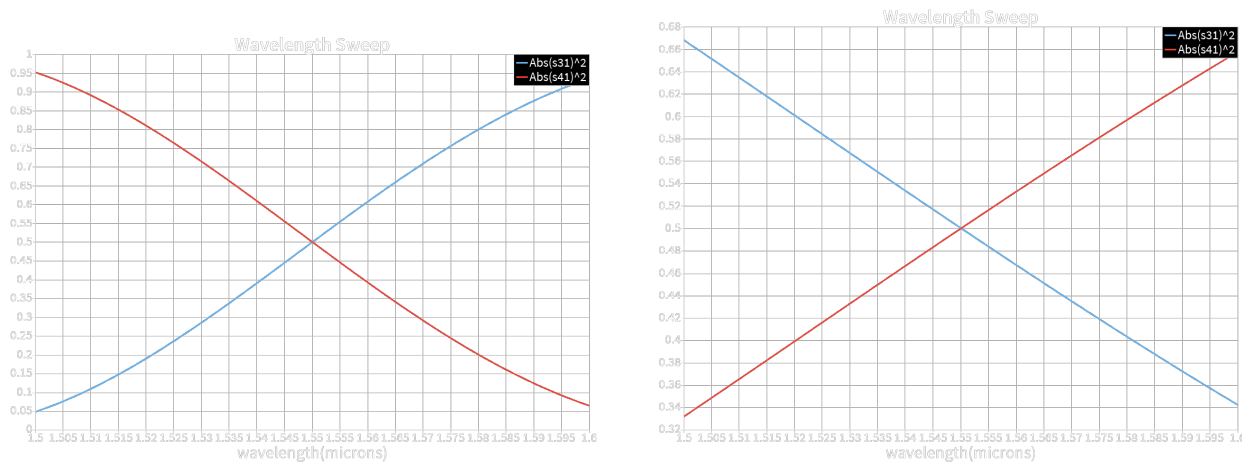
(b) $w = 500$ nm, $g = 150$ nm.

Figure 13: EME coupling-length sweeps for the selected beam-splitter geometries. The plotted quantities are the output powers extracted from the S-matrix, with $|s_{31}|^2$ and $|s_{41}|^2$ corresponding to the two output ports for excitation from physical port 1. The 50/50 operating points occur where the two output powers cross at 0.5.

After selecting a 50/50 coupling length, the wavelength response is simulated to evaluate the bandwidth of the beam splitter. Figure 14 shows the wavelength sweep for the selected coupling-length cases. The plotted curves show how the through and cross output powers vary across the 1500–1600 nm band. The crossing point near

$$\lambda = 1550 \text{ nm}$$

corresponds to the designed 50/50 operating wavelength. Away from this wavelength, the splitting ratio changes because the even–odd supermode index splitting, and therefore the coupling coefficient, is wavelength dependent.

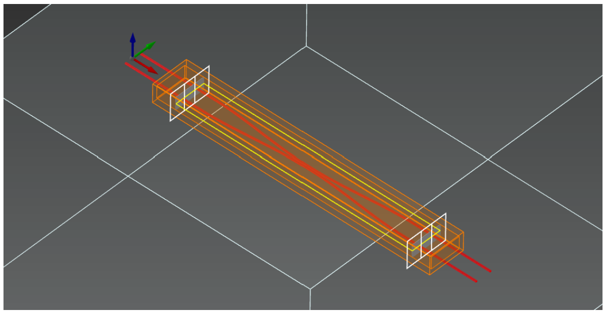


(a) Wavelength sweep for the $w = 300$ nm design.

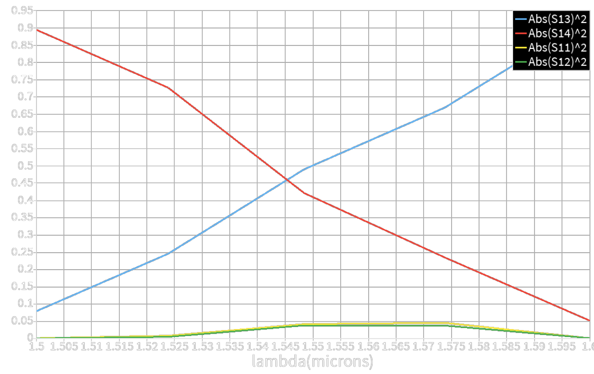
(b) Wavelength sweep for the $w = 500$ nm design.

Figure 14: Wavelength response of the tuned beam splitters. The crossing of $|s_{31}|^2$ and $|s_{41}|^2$ indicates the 50/50 splitting point. The wavelength dependence of the splitting ratio comes from the dispersion of the coupled supermodes.

As a preliminary full-wave check, a quick low-accuracy FDTD simulation was also performed for one of the designed 50/50 beam-splitter geometries. This simulation was not intended as the final quantitative validation, but rather as a sanity check to verify that the full three-dimensional structure shows the expected power-transfer behavior. The FDTD setup includes the imported silicon waveguide geometry, the surrounding oxide cladding region, mode sources/ports at the input and output waveguides, and frequency-domain monitors for extracting the wavelength-dependent S-matrix response. The simulation region, port placement, and extracted S-matrix response are shown in Fig. 15.



(a) Three-dimensional FDTD setup.



(b) Low-accuracy FDTD S-matrix wavelength sweep.

Figure 15: Preliminary low-accuracy FDTD validation of the designed 50/50 beam splitter. (a) The imported GDS geometry is extruded into the silicon device layer, and the FDTD region is placed around the complete routed directional-coupler structure. (b) The two dominant output terms, $|S_{13}|^2$ and $|S_{14}|^2$, become approximately balanced near the design wavelength, showing that the full-wave response is roughly consistent with the expected beam-splitting behavior.

For excitation from the selected input port, the two dominant output terms are approximately balanced near the design wavelength. In this FDTD port convention, the main output channels are represented by $|S_{13}|^2$ and $|S_{14}|^2$, while the reflection and leakage terms remain much smaller. Around the 50/50 point, the two output powers are close to each other, confirming that the device behavior is roughly consistent with the EME tuning result. The residual imbalance and small parasitic terms are expected to be affected by the coarse simulation accuracy, mesh resolution, finite simulation time, and the preliminary port/monitor setup. Therefore, this result is used only as an initial validation of the qualitative behavior, while the final performance should be checked later using a higher-accuracy FDTD run.

These EME and preliminary FDTD results show that the complete routed beam splitter can be tuned to 50/50 operation by adjusting the central coupling length. The optimized value is not necessarily identical to the simple coupled-mode-theory estimate because the full layout includes S-bend transitions, finite port sections, and the actual modal evolution through the routed structure. Therefore, the final coupling length should be selected from the layout-level EME sweep rather than from the isolated straight-coupler calculation alone.

After the EME sweep identifies the best value of L_c, the final tuned design can be verified using a higher-accuracy FDTD simulation. This final verification should use a finer mesh, longer simulation time, and carefully converged port/monitor settings to check the full-wave response of the selected geometry, estimate radiation loss and reflection, and compare the infinite-cladding and finite-cladding models. In this way, the complete design workflow uses coupled-mode theory for the initial physical estimate, EME for efficient layout-level tuning, a quick FDTD run for qualitative behavioral confirmation, and a final converged FDTD simulation for electromagnetic validation.

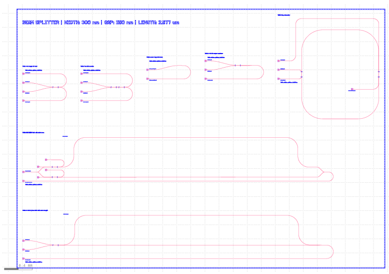
Test Structures

For each selected beam-splitter design point, defined by the waveguide width w, the coupling gap g, and the coupling length L_c, a set of test structures is generated and placed on the final chip. The purpose of these structures is to make the beam splitter measurable in several configurations, rather than relying only on a single four-port measurement. Each test structure uses the same beam-splitter geometry, so that the measured response can be compared across direct transmission, cross coupling, cascaded operation, phase-sensitive interference, and reference routing structures.

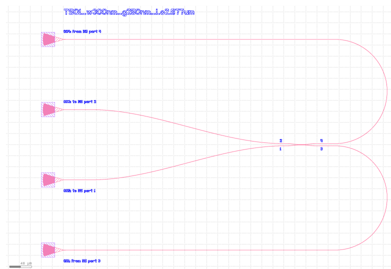
The test structures are designed with grating-coupler access ports and routed waveguides, so that the devices can be measured using a fiber array. The labels on each structure include the beam-splitter parameters,

$$w, \qquad g, \qquad L_c,$$

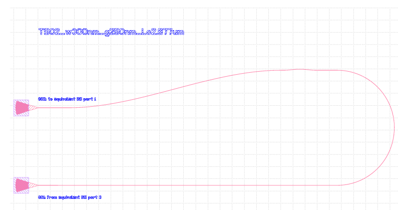
which makes each cell directly identifiable under the microscope and in the final GDS layout.



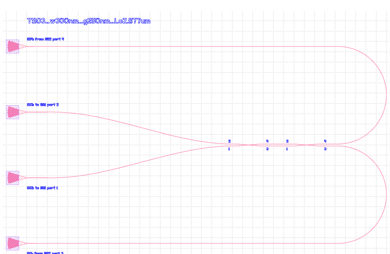
(a) Master cell containing all seven test structures.



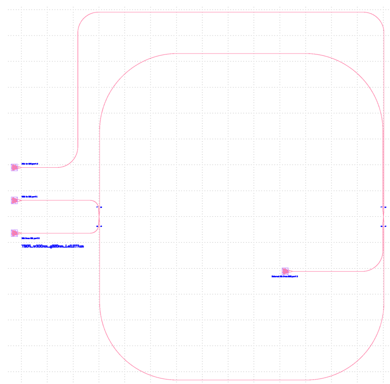
(b) TS01: four-port beam-splitter characterization.



(c) TS02: single-input beam-splitter routing reference.

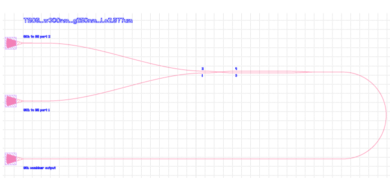


(d) TS03: equivalent-path reference structure.



(e) TS04: cascaded beam-splitter structure.

(f) TS05: three-grating-coupler loop/interference structure.



(g) TS06: beam-splitter combiner structure.

(h) TS07: phase-sensitive MZI-style beam-splitter structure.

Figure 16: Generated test-structure layouts for one beam-splitter design point. The first panel shows the master cell containing all test structures, while the remaining panels show TS01–TS07 individually.

1. **TS01: Four-port beam-splitter characterization.**

TS01 is the main direct characterization structure for the beam splitter. All four ports of the directional coupler are routed to grating couplers. This allows the transmission from each input port to the two output ports to be measured directly. From this structure, the splitting ratio, excess loss, imbalance, and port-to-port symmetry of the beam splitter can be extracted.

2. **TS02: Single-input beam-splitter routing reference.**

   TS02 connects one grating-coupler input to the beam splitter and routes the relevant output ports to grating couplers. This structure is used as a simplified measurement version of the beam splitter, where one input path is emphasized. It helps verify the expected bar and cross transmission of the device with reduced routing complexity compared with the full four-port structure.

3. **TS03: Equivalent-path reference structure.**

   TS03 provides an equivalent routed path using the same general input and output access geometry, but with a simplified beam-splitter connection. This structure is useful for separating the loss of the beam splitter from the loss of the grating couplers, tapers, and long routing waveguides. It therefore acts as a reference for normalizing the measured transmission of the other test structures.

4. **TS04: Cascaded beam-splitter structure.**

   TS04 contains two beam splitters connected in series. This structure tests the behavior of the beam splitter when it is used as a building block in a larger interferometric circuit. By measuring the output powers of the cascaded device, the consistency of the splitting ratio and the accumulated insertion loss can be evaluated. This structure is also useful for checking whether the designed beam splitter behaves correctly when used in multi-stage photonic circuits.

5. **TS05: Three-grating-coupler loop/interference structure.**

   TS05 uses a three-port grating-coupler access configuration with a routed loop section. This structure is designed to test interference behavior and phase accumulation in a compact circuit that includes the beam splitter. Because one of the paths is routed through a longer waveguide loop, this structure can reveal phase-dependent behavior and wavelength-dependent interference effects.

6. **TS06: Beam-splitter combiner structure.**

   TS06 is designed to test the beam splitter in a combining configuration. Two input grating couplers excite the two input arms, and the output is routed to a combined output port. This structure is useful for checking coherent combination, relative phase sensitivity, and the behavior of the beam splitter when it is operated in the reverse direction compared with the standard splitting measurement.

7. **TS07: Phase-sensitive MZI-style beam-splitter structure.**

   TS07 is a phase-sensitive structure based on beam-splitter operation in an interferometric layout. It includes an additional path length in one arm, so the output becomes wavelength dependent. This makes the structure useful for observing interference fringes, estimating relative phase accumulation, and checking whether the beam splitter can be used reliably inside Mach–Zehnder-type circuits.

After generating the seven individual test structures, they are gathered into a single master cell, as shown in Fig. 16a. The master cell places TS01–TS07 inside a common bounding box and adds a large label containing the beam-splitter design parameters,

$$\text{WIDTH} = w, \qquad \text{GAP} = g, \qquad \text{LENGTH} = L_{\text{c}}.$$

This boxed master cell is then used as the repeated unit in the final chip layout. By changing only the beam-splitter width, gap, and coupling length, the same set of seven diagnostic structures can be generated for every design point in the sweep.

DOE for Coupling-Length Sweep

To experimentally identify the optimal 50/50 beam-splitter condition, a design-of-experiment (DOE) layout is generated by sweeping the coupling length around the value predicted by simulation. Although the electromagnetic simulation provides an initial estimate for the required coupling length, the fabricated device can deviate from this value because of lithography bias, etch variation, sidewall roughness, and uncertainty in the material and geometry models. Therefore, several beam splitters with different coupling lengths are included on the final chip.

The DOE is arranged as a 4×8 array of master cells. Each master cell contains the full set of test structures described in Sec. 2, including TS01–TS07, for one specific beam-splitter design point. In this way, every coupling-length value can be tested through the same measurement structures, allowing a direct comparison between devices.

The first two rows of the DOE array correspond to the $w = 300$ nm beam-splitter design, while the last two rows correspond to the $w = 500$ nm design. For each waveguide width, two different coupling-length sweeps are included: a fine sweep and a coarse sweep.

For each width, the first row is a fine detuning sweep around the simulated 50/50 coupling length,

$$L_{\text{sim}}.$$

The coupling length is varied in small steps before and after L_{sim}, with a step size chosen on the order of

$$\Delta L_{\text{fine}} \approx \frac{L_\pi}{10},$$

where L_π is the full power-transfer beat length of the directional coupler. This fine sweep is intended to resolve the optimum 50/50 point with high accuracy and to compensate for small systematic shifts between the simulated and fabricated devices.

The second row for each width is a coarse coupling-length sweep. In this row, the coupling length starts from the simulated value L_{sim} and is increased up to an extended value of approximately

$$L_{\max} \approx 1.3 L_\pi.$$

This wider sweep is included to observe the full coupling oscillation trend and to ensure that the design captures the 50/50 condition even if the fabricated coupling strength differs significantly from the simulated value. The coarse sweep also provides information about the coupling period, fabrication-induced shifts, and possible excess loss at longer interaction lengths.

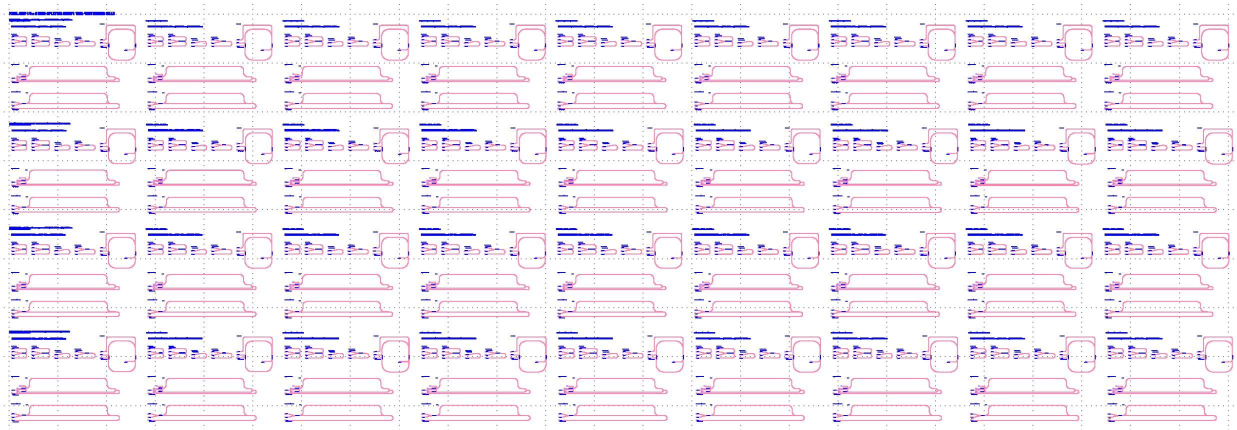


Figure 17: Final-chip DOE layout for sweeping the beam-splitter coupling length. The array contains 4×8 master cells. The first two rows correspond to the $w = 300$ nm design, and the last two rows correspond to the $w = 500$ nm design. For each width, the first row is a fine sweep around the simulated 50/50 length, while the second row is a coarse sweep from the simulated length up to approximately $1.3L_\pi$.

Overall, this DOE structure provides both local and global information about the coupling-length dependence of the beam splitter. The fine sweep is used to select the best fabricated 50/50 device, while the coarse sweep verifies the coupling trend over a larger length range and provides a fabrication-tolerant backup in case the simulated coupling length is shifted.

Discussion

In this work, the beam-splitter design was developed through a complete workflow starting from single-waveguide mode analysis, followed by double-waveguide supermode simulations, coupling-length estimation, GDS layout generation, and final-chip test-structure organization. The main goal was to design compact directional-coupler beam splitters on a 220 nm SOI platform and to include enough test structures on the final chip to experimentally identify the best 50/50 device after fabrication.

Several design choices were made to balance compactness, robustness, and measurement practicality. The waveguide widths of 300 nm and 500 nm were selected because they represent two different coupling regimes. The narrower waveguide has a larger evanescent tail and can therefore achieve coupling at larger gaps, while the wider waveguide is more strongly confined and is more compatible with compact routing, but requires smaller gaps to obtain strong coupling. For each design, the coupling gap and coupling length were chosen from the double-waveguide simulations, and additional length sweeps were included on the final chip to compensate for fabrication uncertainty.

The final layout was organized as a design-of-experiment array. For each selected beam-splitter design point, seven test structures were generated. These structures allow the same beam splitter to be measured in direct four-port, reference, cascaded, combiner, and interferometric configurations. This is important because a single transmission measurement is usually not enough to fully understand the behavior of a fabricated beam splitter. The test structures provide additional information about insertion loss, splitting ratio, routing loss, phase sensitivity, wavelength dependence, and the effect of cascading.

There are several additional tasks that would make the work more complete, but they were not finished in the present design cycle. First, a full high-accuracy FDTD simulation of the final optimized beam splitter should be performed over the complete wavelength range. In this work, FDTD was mainly used as a quick verification step with reduced accuracy to confirm the general behavior of the designed 50/50 structure. A more accurate simulation would give a better estimate of radiation loss, back-reflection, finite-cladding effects, and parasitic scattering.

Second, a more complete EME-based wavelength sweep should be performed for all final design candidates. The EME method is more efficient than FDTD for long directional-coupler structures and would allow a fast extraction of the full four-port S-matrix versus wavelength. This would be useful for evaluating the bandwidth of the 50/50 splitting ratio and the relative phase between the output ports.

Third, fabrication-aware simulations should be added. In particular, the waveguide width, coupling gap, sidewall angle, etch depth, and layer thickness should be swept around their nominal values. This would quantify the sensitivity of the splitting ratio and phase response to fabrication errors and would make the design more predictive before tape-out.

Fourth, the grating couplers and edge/fiber coupling structures were treated as standard access components rather than being fully optimized in this workflow. A complete chip-level design would include simulations or measured calibration structures for the grating couplers, tapers, and routing waveguides. This would allow the beam-splitter loss to be separated more accurately from the coupling and routing losses.

Finally, after fabrication, the DOE structures should be measured systematically. The fine coupling-length sweep should be used to identify the best fabricated 50/50 beam splitter, while the coarse sweep should be used to extract the experimental coupling period and compare it with the simulated value. The reference and interferometric structures should then be used to estimate excess loss, imbalance, wavelength dependence, and phase behavior. These measurements would provide the final validation of the beam-splitter design and would indicate which geometry should be used in future integrated photonic circuits.